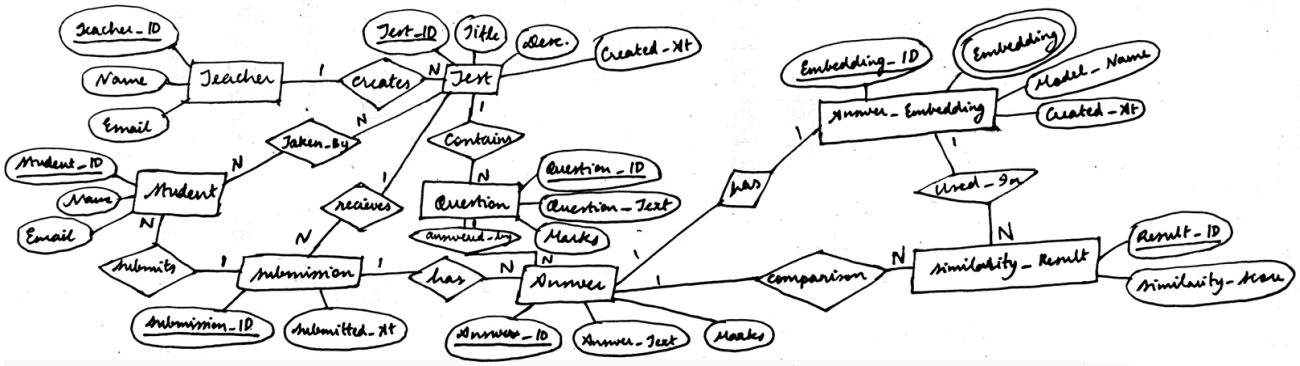
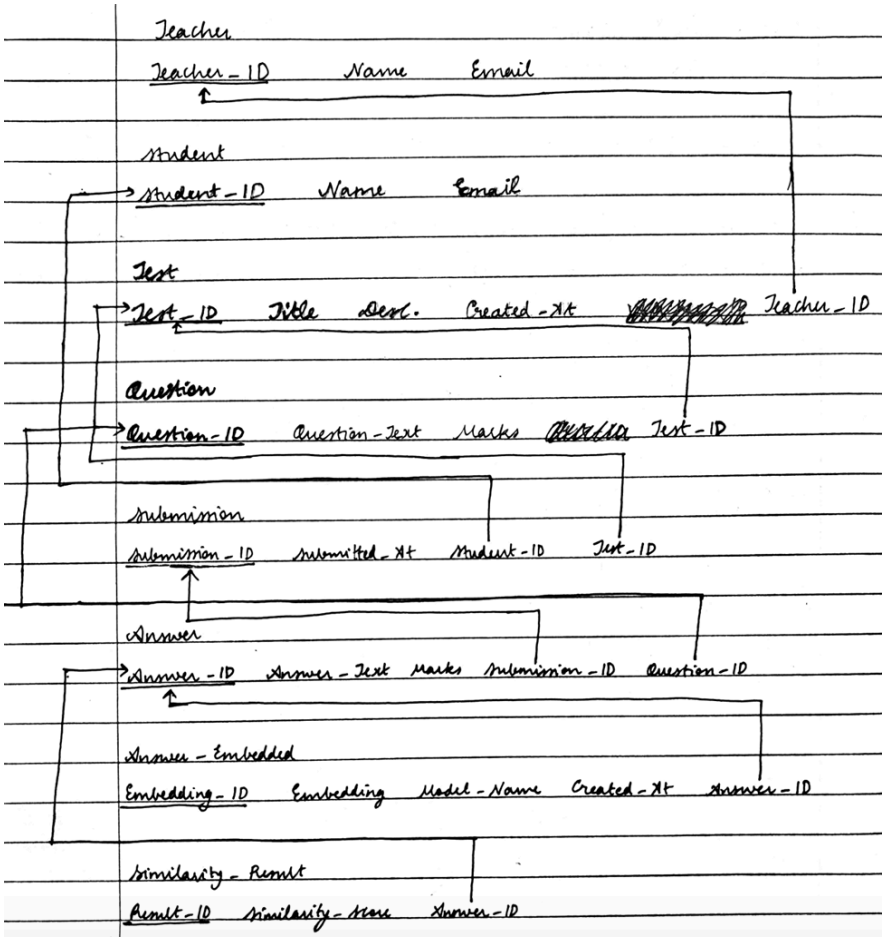
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DBMS Lab 4

Project Title: Test/Quiz system with Similarity Detection between Answers.
Database Requirement: SQL + Vector Database
Database used: PostgreSQL
Database Design:
ER Diagram:



Relational Schema:



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This project uses PostgreSQL with the pgvector extension to handle the vector part of the system. When a student submits an answer, the answer is converted into a vector (embedding) that represents its meaning. These embeddings are stored along with details such as the Answer_ID and Question_ID. When another answer is submitted, the system compares its embedding with other answers to the same question and finds the most similar ones using vector similarity search. The similarity score can then be used to identify answers that have similar meanings.

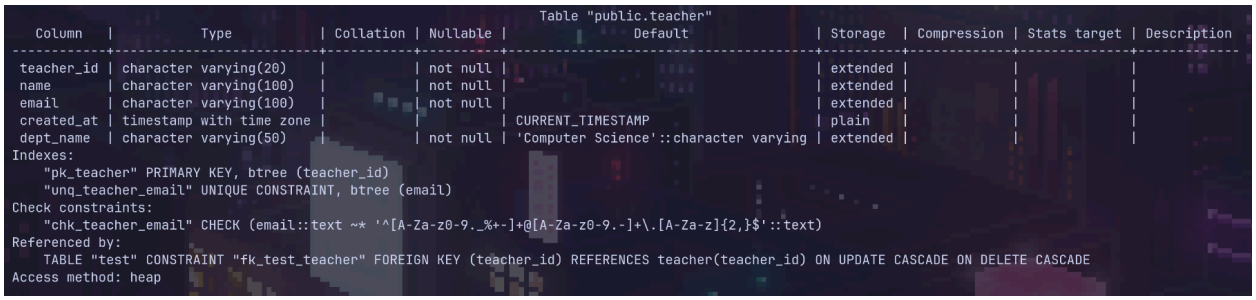
SQL Tables

List of Relations:-

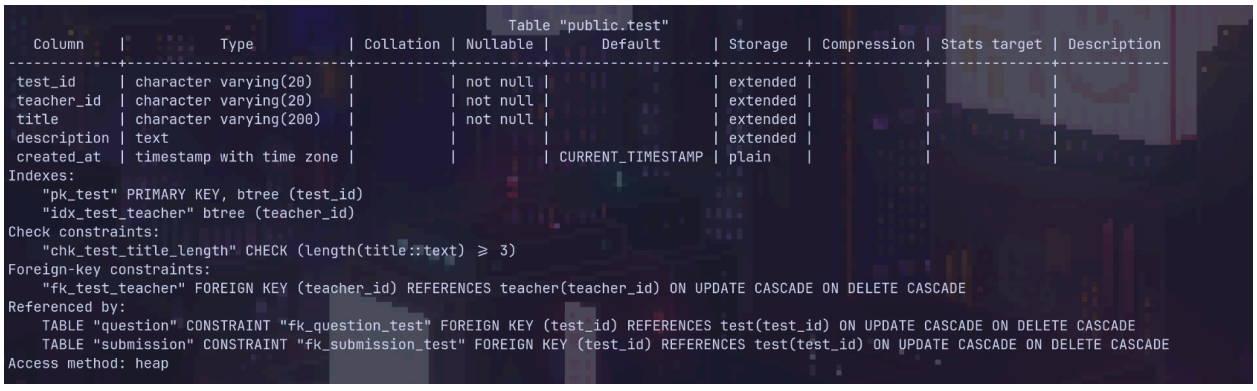


```
                                List of relations
 Schema |       Name         | Type  |  Owner   | Persistence | Access method |    Size    | Description
--------+--------------------+-------+----------+-------------+---------------+------------+-------------
 public | answer             | table | postgres | permanent   | heap          | 8192 bytes |
 public | answer_embedding   | table | postgres | permanent   | heap          | 8192 bytes |
 public | question           | table | postgres | permanent   | heap          | 8192 bytes |
 public | similarity_result  | table | postgres | permanent   | heap          | 0 bytes    |
 public | student            | table | postgres | permanent   | heap          | 0 bytes    |
 public | submission         | table | postgres | permanent   | heap          | 0 bytes    |
 public | teacher            | table | postgres | permanent   | heap          | 0 bytes    |
 public | test               | table | postgres | permanent   | heap          | 8192 bytes |
```

Tables:-

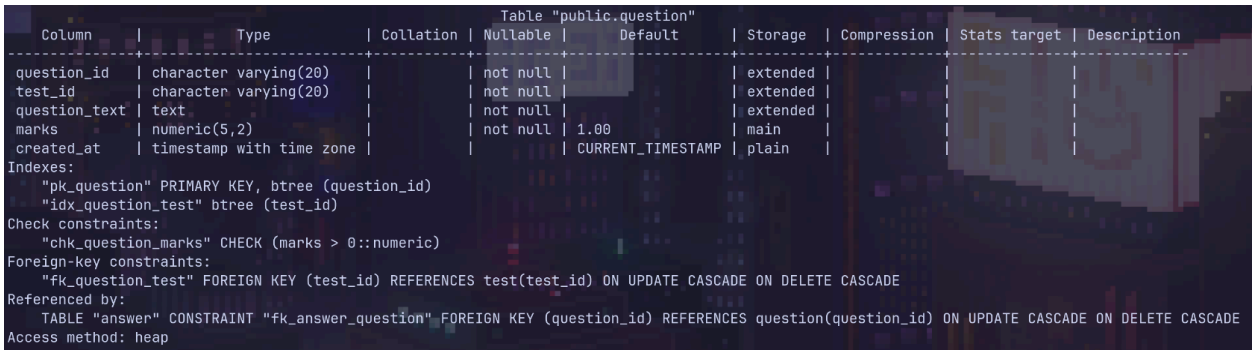


```
                                        Table "public.teacher"
   Column   |          Type          | Collation | Nullable |              Default               | Storage  | Compression | Stats target | Description
------------+------------------------+-----------+----------+------------------------------------+----------+-------------+--------------+-------------
 teacher_id | character varying(20)  |           | not null |                                    | extended |             |              |
 name       | character varying(100) |           | not null |                                    | extended |             |              |
 email      | character varying(100) |           | not null |                                    | extended |             |              |
 created_at | timestamp with time zone |         |          | CURRENT_TIMESTAMP                  | plain    |             |              |
 dept_name  | character varying(50)  |           | not null | 'Computer Science'::character varying | extended |          |              |
Indexes:
    "pk_teacher" PRIMARY KEY, btree (teacher_id)
    "unq_teacher_email" UNIQUE CONSTRAINT, btree (email)
Check constraints:
    "chk_teacher_email" CHECK (email::text ~* '^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$'::text)
Referenced by:
    TABLE "test" CONSTRAINT "fk_test_teacher" FOREIGN KEY (teacher_id) REFERENCES teacher(teacher_id) ON UPDATE CASCADE ON DELETE CASCADE
Access method: heap
```

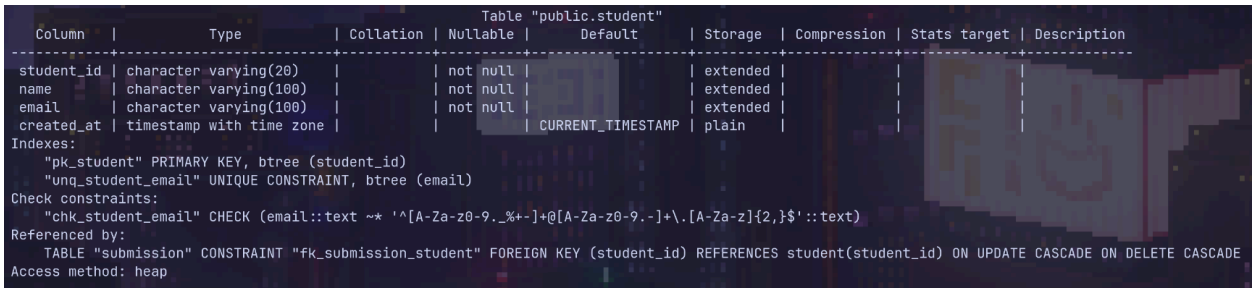


```
                                          Table "public.test"
   Column    |          Type          | Collation | Nullable |      Default      | Storage  | Compression | Stats target | Description
-------------+------------------------+-----------+----------+-------------------+----------+-------------+--------------+-------------
 test_id     | character varying(20)  |           | not null |                   | extended |             |              |
 teacher_id  | character varying(20)  |           | not null |                   | extended |             |              |
 title       | character varying(200) |           | not null |                   | extended |             |              |
 description | text                   |           |          |                   | extended |             |              |
 created_at  | timestamp with time zone |         |          | CURRENT_TIMESTAMP | plain    |             |              |
Indexes:
    "pk_test" PRIMARY KEY, btree (test_id)
    "idx_test_teacher" btree (teacher_id)
Check constraints:
    "chk_test_title_length" CHECK (length(title::text) >= 3)
Foreign-key constraints:
    "fk_test_teacher" FOREIGN KEY (teacher_id) REFERENCES teacher(teacher_id) ON UPDATE CASCADE ON DELETE CASCADE
Referenced by:
    TABLE "question" CONSTRAINT "fk_question_test" FOREIGN KEY (test_id) REFERENCES test(test_id) ON UPDATE CASCADE ON DELETE CASCADE
    TABLE "submission" CONSTRAINT "fk_submission_test" FOREIGN KEY (test_id) REFERENCES test(test_id) ON UPDATE CASCADE ON DELETE CASCADE
Access method: heap
```

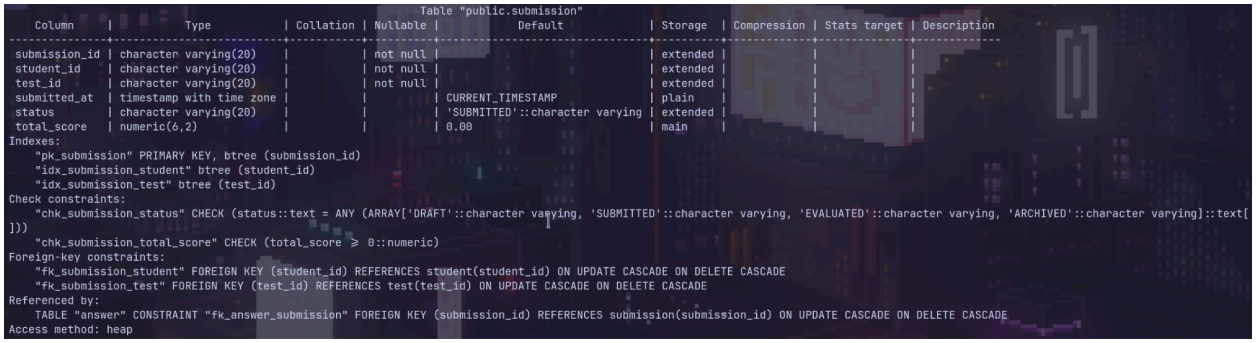
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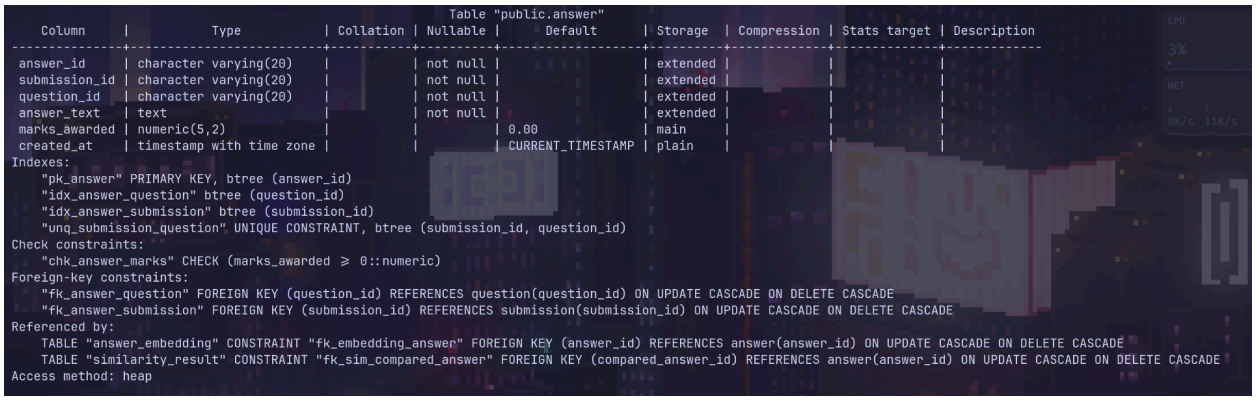
```
                                                Table "public.question"
    Column     |           Type           | Collation | Nullable |      Default      | Storage  | Compression | Stats target | Description
---------------+--------------------------+-----------+----------+-------------------+----------+-------------+--------------+-------------
 question_id   | character varying(20)    |           | not null |                   | extended |             |              |
 test_id       | character varying(20)    |           | not null |                   | extended |             |              |
 question_text | text                     |           | not null |                   | extended |             |              |
 marks         | numeric(5,2)             |           | not null | 1.00              | main     |             |              |
 created_at    | timestamp with time zone |           |          | CURRENT_TIMESTAMP | plain    |             |              |
Indexes:
    "pk_question" PRIMARY KEY, btree (question_id)
    "idx_question_test" btree (test_id)
Check constraints:
    "chk_question_marks" CHECK (marks > 0::numeric)
Foreign-key constraints:
    "fk_question_test" FOREIGN KEY (test_id) REFERENCES test(test_id) ON UPDATE CASCADE ON DELETE CASCADE
Referenced by:
    TABLE "answer" CONSTRAINT "fk_answer_question" FOREIGN KEY (question_id) REFERENCES question(question_id) ON UPDATE CASCADE ON DELETE CASCADE
Access method: heap
```



```
                                                Table "public.student"
   Column   |           Type           | Collation | Nullable |      Default      | Storage  | Compression | Stats target | Description
------------+--------------------------+-----------+----------+-------------------+----------+-------------+--------------+-------------
 student_id | character varying(20)    |           | not null |                   | extended |             |              |
 name       | character varying(100)   |           | not null |                   | extended |             |              |
 email      | character varying(100)   |           | not null |                   | extended |             |              |
 created_at | timestamp with time zone |           |          | CURRENT_TIMESTAMP | plain    |             |              |
Indexes:
    "pk_student" PRIMARY KEY, btree (student_id)
    "unq_student_email" UNIQUE CONSTRAINT, btree (email)
Check constraints:
    "chk_student_email" CHECK (email::text ~* '^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$'::text)
Referenced by:
    TABLE "submission" CONSTRAINT "fk_submission_student" FOREIGN KEY (student_id) REFERENCES student(student_id) ON UPDATE CASCADE ON DELETE CASCADE
Access method: heap
```



```
                                                Table "public.submission"
    Column     |           Type           | Collation | Nullable |           Default            | Storage  | Compression | Stats target | Description
---------------+--------------------------+-----------+----------+------------------------------+----------+-------------+--------------+-------------
 submission_id | character varying(20)    |           | not null |                              | extended |             |              |
 student_id    | character varying(20)    |           | not null |                              | extended |             |              |
 test_id       | character varying(20)    |           | not null |                              | extended |             |              |
 submitted_at  | timestamp with time zone |           |          | CURRENT_TIMESTAMP            | plain    |             |              |
 status        | character varying(20)    |           |          | 'SUBMITTED'::character varying | extended |             |              |
 total_score   | numeric(6,2)             |           |          | 0.00                         | main     |             |              |
Indexes:
    "pk_submission" PRIMARY KEY, btree (submission_id)
    "idx_submission_student" btree (student_id)
    "idx_submission_test" btree (test_id)
Check constraints:
    "chk_submission_status" CHECK (status::text = ANY (ARRAY['DRAFT'::character varying, 'SUBMITTED'::character varying, 'EVALUATED'::character varying, 'ARCHIVED'::character varying]::text[
]))
    "chk_submission_total_score" CHECK (total_score >= 0::numeric)
Foreign-key constraints:
    "fk_submission_student" FOREIGN KEY (student_id) REFERENCES student(student_id) ON UPDATE CASCADE ON DELETE CASCADE
    "fk_submission_test" FOREIGN KEY (test_id) REFERENCES test(test_id) ON UPDATE CASCADE ON DELETE CASCADE
Referenced by:
    TABLE "answer" CONSTRAINT "fk_answer_submission" FOREIGN KEY (submission_id) REFERENCES submission(submission_id) ON UPDATE CASCADE ON DELETE CASCADE
Access method: heap
```



```
                                                Table "public.answer"
    Column     |           Type           | Collation | Nullable |      Default      | Storage  | Compression | Stats target | Description
---------------+--------------------------+-----------+----------+-------------------+----------+-------------+--------------+-------------
 answer_id     | character varying(20)    |           | not null |                   | extended |             |              |
 submission_id | character varying(20)    |           | not null |                   | extended |             |              |
 question_id   | character varying(20)    |           | not null |                   | extended |             |              |
 answer_text   | text                     |           | not null |                   | extended |             |              |
 marks_awarded | numeric(5,2)             |           |          | 0.00              | main     |             |              |
 created_at    | timestamp with time zone |           |          | CURRENT_TIMESTAMP | plain    |             |              |
Indexes:
    "pk_answer" PRIMARY KEY, btree (answer_id)
    "idx_answer_question" btree (question_id)
    "idx_answer_submission" btree (submission_id)
    "unq_submission_question" UNIQUE CONSTRAINT, btree (submission_id, question_id)
Check constraints:
    "chk_answer_marks" CHECK (marks_awarded >= 0::numeric)
Foreign-key constraints:
    "fk_answer_question" FOREIGN KEY (question_id) REFERENCES question(question_id) ON UPDATE CASCADE ON DELETE CASCADE
    "fk_answer_submission" FOREIGN KEY (submission_id) REFERENCES submission(submission_id) ON UPDATE CASCADE ON DELETE CASCADE
Referenced by:
    TABLE "answer_embedding" CONSTRAINT "fk_embedding_answer" FOREIGN KEY (answer_id) REFERENCES answer(answer_id) ON UPDATE CASCADE ON DELETE CASCADE
    TABLE "similarity_result" CONSTRAINT "fk_sim_compared_answer" FOREIGN KEY (compared_answer_id) REFERENCES answer(answer_id) ON UPDATE CASCADE ON DELETE CASCADE
Access method: heap
```

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```
                                                Table "public.answer_embedding"
   Column    |          Type          | Collation | Nullable |                    Default                     | Storage  | Compression | Stats target | Description
-------------+------------------------+-----------+----------+------------------------------------------------+----------+-------------+--------------+-------------
 embedding_id | character varying(20)  |           | not null |                                                | extended |             |              |
 answer_id    | character varying(20)  |           | not null |                                                | extended |             |              |
 embedding    | vector(384)            |           | not null |                                                | external |             |              |
 model_name   | character varying(100) |           | not null | 'sentence-transformers/all-MiniLM-L6-v2'::character varying | extended |             |              |
 created_at   | timestamp with time zone |         |          | CURRENT_TIMESTAMP                              | plain    |             |              |
Indexes:
    "pk_answer_embedding" PRIMARY KEY, btree (embedding_id)
    "idx_embedding_vector_hnsw" hnsw (embedding vector_cosine_ops)
    "unq_embedding_answer" UNIQUE CONSTRAINT, btree (answer_id)
Foreign-key constraints:
    "fk_embedding_answer" FOREIGN KEY (answer_id) REFERENCES answer(answer_id) ON UPDATE CASCADE ON DELETE CASCADE
Referenced by:
    TABLE "similarity_result" CONSTRAINT "fk_sim_embedding" FOREIGN KEY (embedding_id) REFERENCES answer_embedding(embedding_id) ON UPDATE CASCADE ON DELETE CASCADE
Access method: heap
```

```
                                          Table "public.similarity_result"
      Column       |          Type          | Collation | Nullable |      Default      | Storage  | Compression | Stats target | Description
-------------------+------------------------+-----------+----------+-------------------+----------+-------------+--------------+-------------
 result_id         | character varying(20)  |           | not null |                   | extended |             |              |
 embedding_id      | character varying(20)  |           | not null |                   | extended |             |              |
 compared_answer_id | character varying(20) |           | not null |                   | extended |             |              |
 similarity_score  | numeric(5,4)           |           | not null |                   | main     |             |              |
 evaluated_at      | timestamp with time zone |         |          | CURRENT_TIMESTAMP | plain    |             |              |
Indexes:
    "pk_similarity_result" PRIMARY KEY, btree (result_id, embedding_id)
    "idx_sim_compared_answer" btree (compared_answer_id)
Check constraints:
    "chk_similarity_score" CHECK (similarity_score >= 0.0000 AND similarity_score <= 1.0000)
Foreign-key constraints:
    "fk_sim_compared_answer" FOREIGN KEY (compared_answer_id) REFERENCES answer(answer_id) ON UPDATE CASCADE ON DELETE CASCADE
    "fk_sim_embedding" FOREIGN KEY (embedding_id) REFERENCES answer_embedding(embedding_id) ON UPDATE CASCADE ON DELETE CASCADE
Access method: heap
```

List of Indexes:-

```
                                                List of relations
 Schema |          Name           | Type  |  Owner   |       Table       | Persistence | Access method |    Size    | Description
--------+-------------------------+-------+----------+-------------------+-------------+---------------+------------+-------------
 public | idx_answer_question     | index | postgres | answer            | permanent   | btree         | 8192 bytes |
 public | idx_answer_submission   | index | postgres | answer            | permanent   | btree         | 8192 bytes |
 public | idx_embedding_vector_hnsw | index | postgres | answer_embedding | permanent   | hnsw          | 16 kB      |
 public | idx_question_test       | index | postgres | question          | permanent   | btree         | 8192 bytes |
 public | idx_sim_compared_answer | index | postgres | similarity_result | permanent   | btree         | 8192 bytes |
 public | idx_submission_student  | index | postgres | submission        | permanent   | btree         | 8192 bytes |
 public | idx_submission_test     | index | postgres | submission        | permanent   | btree         | 8192 bytes |
 public | idx_test_teacher        | index | postgres | test              | permanent   | btree         | 8192 bytes |
 public | pk_answer               | index | postgres | answer            | permanent   | btree         | 8192 bytes |
 public | pk_answer_embedding     | index | postgres | answer_embedding  | permanent   | btree         | 8192 bytes |
 public | pk_question             | index | postgres | question          | permanent   | btree         | 8192 bytes |
 public | pk_similarity_result    | index | postgres | similarity_result | permanent   | btree         | 8192 bytes |
 public | pk_student              | index | postgres | student           | permanent   | btree         | 8192 bytes |
 public | pk_submission           | index | postgres | submission        | permanent   | btree         | 8192 bytes |
 public | pk_teacher              | index | postgres | teacher           | permanent   | btree         | 8192 bytes |
 public | pk_test                 | index | postgres | test              | permanent   | btree         | 8192 bytes |
 public | unq_embedding_answer    | index | postgres | answer_embedding  | permanent   | btree         | 8192 bytes |
 public | unq_student_email       | index | postgres | student           | permanent   | btree         | 8192 bytes |
 public | unq_submission_question | index | postgres | answer            | permanent   | btree         | 8192 bytes |
 public | unq_teacher_email       | index | postgres | teacher           | permanent   | btree         | 8192 bytes |
(20 rows)
```

List of views on Postgres:-

```
                                List of relations
 Schema |             Name              | Type |  Owner   | Persistence |  Size   | Description
--------+-------------------------------+------+----------+-------------+---------+-------------
 public | v_similarity_evaluation       | view | postgres | permanent   | 0 bytes |
 public | v_student_submission_summary  | view | postgres | permanent   | 0 bytes |
 public | v_test_details                | view | postgres | permanent   | 0 bytes |
(3 rows)
```

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Commands run

```
-- 1.1 TEACHER Table
CREATE TABLE TEACHER (
    teacher_id VARCHAR(20) NOT NULL,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100) NOT NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT pk_teacher PRIMARY KEY (teacher_id),
    CONSTRAINT unq_teacher_email UNIQUE (email),
    CONSTRAINT chk_teacher_email CHECK (email ~*
'^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$')
);

-- 1.2 TEST Table
CREATE TABLE TEST (
    test_id VARCHAR(20) NOT NULL,
    teacher_id VARCHAR(20) NOT NULL,
    title VARCHAR(200) NOT NULL,
    description TEXT,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT pk_test PRIMARY KEY (test_id),
    CONSTRAINT fk_test_teacher FOREIGN KEY (teacher_id)
        REFERENCES TEACHER(teacher_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);

-- 1.3 QUESTION Table

CREATE TABLE QUESTION (
    question_id VARCHAR(20) NOT NULL,
    test_id VARCHAR(20) NOT NULL,
    question_text TEXT NOT NULL,
    marks NUMERIC(5, 2) NOT NULL DEFAULT 1.00,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT pk_question PRIMARY KEY (question_id),
    CONSTRAINT fk_question_test FOREIGN KEY (test_id)
        REFERENCES TEST(test_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    CONSTRAINT chk_question_marks CHECK (marks > 0)
);

-- 1.4 STUDENT Table
CREATE TABLE STUDENT (
```

```
    student_id VARCHAR(20) NOT NULL,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100) NOT NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT pk_student PRIMARY KEY (student_id),
    CONSTRAINT unq_student_email UNIQUE (email),
    CONSTRAINT chk_student_email CHECK (email ~*
'^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$')
);

-- 1.5 SUBMISSION Table
CREATE TABLE SUBMISSION (
    submission_id VARCHAR(20) NOT NULL,
    student_id VARCHAR(20) NOT NULL,
    test_id VARCHAR(20) NOT NULL,
    submitted_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    status VARCHAR(20) DEFAULT 'SUBMITTED',
    total_score NUMERIC(6, 2) DEFAULT 0.00,
    CONSTRAINT pk_submission PRIMARY KEY (submission_id),
    CONSTRAINT fk_submission_student FOREIGN KEY (student_id)
        REFERENCES STUDENT(student_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    CONSTRAINT fk_submission_test FOREIGN KEY (test_id)
        REFERENCES TEST(test_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    CONSTRAINT chk_submission_status CHECK (status IN ('DRAFT', 'SUBMITTED',
'EVALUATED', 'ARCHIVED')),
    CONSTRAINT chk_submission_total_score CHECK (total_score >= 0)
);

-- 1.6 ANSWER Table
CREATE TABLE ANSWER (
    answer_id VARCHAR(20) NOT NULL,
    submission_id VARCHAR(20) NOT NULL,
    question_id VARCHAR(20) NOT NULL,
    answer_text TEXT NOT NULL,
    marks_awarded NUMERIC(5, 2) DEFAULT 0.00,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT pk_answer PRIMARY KEY (answer_id),
    CONSTRAINT fk_answer_submission FOREIGN KEY (submission_id)
        REFERENCES SUBMISSION(submission_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    CONSTRAINT fk_answer_question FOREIGN KEY (question_id)
```

```
        REFERENCES QUESTION(question_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    CONSTRAINT chk_answer_marks CHECK (marks_awarded >= 0),
    CONSTRAINT unq_submission_question UNIQUE (submission_id, question_id)
);

-- 1.7 ANSWER_EMBEDDING Table (using pgvector)
CREATE TABLE ANSWER_EMBEDDING (
    embedding_id VARCHAR(20) NOT NULL,
    answer_id VARCHAR(20) NOT NULL,
    embedding vector(384) NOT NULL,
    model_name VARCHAR(100) NOT NULL DEFAULT 'sentence-transformers/all-MiniLM-L6-v2',
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT pk_answer_embedding PRIMARY KEY (embedding_id),
    CONSTRAINT fk_embedding_answer FOREIGN KEY (answer_id)
        REFERENCES ANSWER(answer_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    CONSTRAINT unq_embedding_answer UNIQUE (answer_id)
);

-- 1.8 SIMILARITY_RESULT Table (Weak Entity)
CREATE TABLE SIMILARITY_RESULT (
    result_id VARCHAR(20) NOT NULL,
    embedding_id VARCHAR(20) NOT NULL,
    compared_answer_id VARCHAR(20) NOT NULL,
    similarity_score NUMERIC(5, 4) NOT NULL,
    evaluated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT pk_similarity_result PRIMARY KEY (result_id, embedding_id),
    CONSTRAINT fk_sim_embedding FOREIGN KEY (embedding_id)
        REFERENCES ANSWER_EMBEDDING(embedding_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    CONSTRAINT fk_sim_compared_answer FOREIGN KEY (compared_answer_id)
        REFERENCES ANSWER(answer_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    CONSTRAINT chk_similarity_score CHECK (similarity_score >= 0.0000 AND
similarity_score <= 1.0000)
);
```